

1 MACHINE FILTER: two boards

Two boards, one module

one KiCad project; the PCB file holds both boards side by side, cut apart for fabrication by tools/separate.sh (KiKit)

MAIN board

JLC SMD assembly, 70 x 100 mm. Back: Daisy Seed3 on sockets, power entry, CV ADC stages U1/U2, audio U3/U4, −10 V reference U5, expansion header J14, optional microSD J15 (DNP). Front: male headers JB1 (15 pins), JB2 (13 pins), JB3 (10 pins).

CONTROL board

Hand-soldered, 70 x 107 mm. Front: 12 jacks, 9 pots, 9 LEDs. Back: pot multiplexer U6, LED drivers U7/U8 (SOIC), through-hole resistors and capacitors, female headers JA1, JA2, JA3.

How they meet

JAn plugs onto JBn, pin k to pin k, about 11 mm apart. Every net on the control board is its own copper, so it has its own name: the main board's name with the prefix CTL_ (CTL_GND, CTL_+12V, CTL_CV_BASE, ...). CTL_x meets x only at the header pins (pages 7 and 12), as in the hardware. tools/check_netlist.py checks that the boards joined there are exactly the one-circuit SKiDL netlist (machine_filter.py).

MAIN board (JLC SMD assembly)

Main: power

page 2

Eurorack power entry
Seed3 VIN filter
−10 V reference

File: main-power.kicad_sch

Main: Daisy Seed3

page 3

Daisy Seed3 (A1)
expansion header J14
clip-LED resistor R70
microSD J15 (optional)

File: main-seed3.kicad_sch

Main: CV inputs 1–4

page 4

BASE, WIDTH, HP RES, LP RES
ADC stages (U2)

File: main-cv-inputs-1-4.kicad_sch

Main: CV inputs 5–8

page 5

EQ FREQ, EQ GAIN, DIST, SMPL RATE
ADC stages (U1)

File: main-cv-inputs-5-8.kicad_sch

Main: audio

page 6

IN L / IN R → U3 → codec
codec → U4 → OUT L / OUT R

File: main-audio.kicad_sch

Main: connector

page 7

JB1, JB2, JB3
to the control board

File: main-connector.kicad_sch

CONTROL board (hand-soldered)

Control: jacks

page 8

12 jacks
clip LED D1

File: control-jacks.kicad_sch

Control: pots

page 9

9 pots
pot multiplexer U6

File: control-pots.kicad_sch

Control: CV LEDs 1–4

page 10

BASE, WIDTH, HP RES, LP RES
LED drivers (U7)

File: control-cv-leds-1-4.kicad_sch

Control: CV LEDs 5–8

page 11

EQ FREQ, EQ GAIN, DIST, SMPL RATE
LED drivers (U8)

File: control-cv-leds-5-8.kicad_sch

Control: connector

page 12

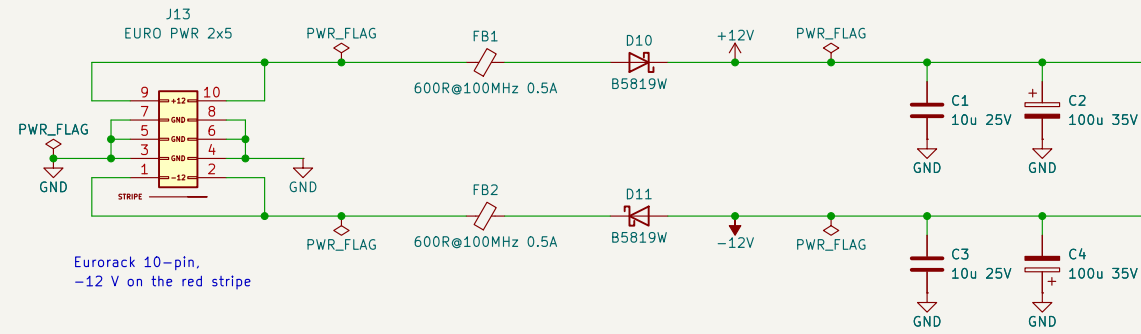
JA1, JA2, JA3
to the main board

File: control-connector.kicad_sch

2 Main board: power entry, Seed3 supply, −10 V reference

Power entry

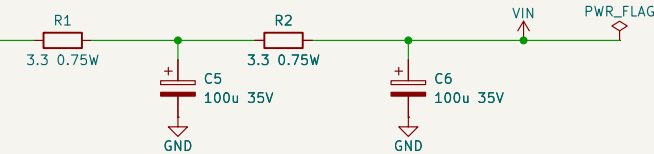
Eurorack header, filters, reverse-polarity diodes, bulk caps



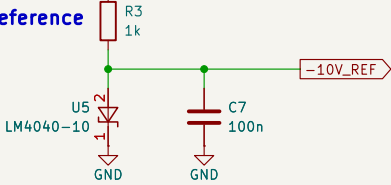
D10, D11: reverse-polarity protection
FB1, FB2: 600 R at 100 MHz, 1 A

Seed3 supply

+12 V → Seed3 VIN, as the Dev Kit



−10 V reference



R3 1k, not the Dev Kit's 2k:
eight 120k offsets draw 0.67 mA

Rails

+12V, −12V

audio op-amps U3, U4 (page 6); to the control board's LED drivers U7, U8 (connector, page 7)

VIN

Seed3 supply input (page 3)

+3V3_A

from the Seed3's 3V3_A pin: CV op-amps U1, U2; to the control board's pots and multiplexer

+3V3_D

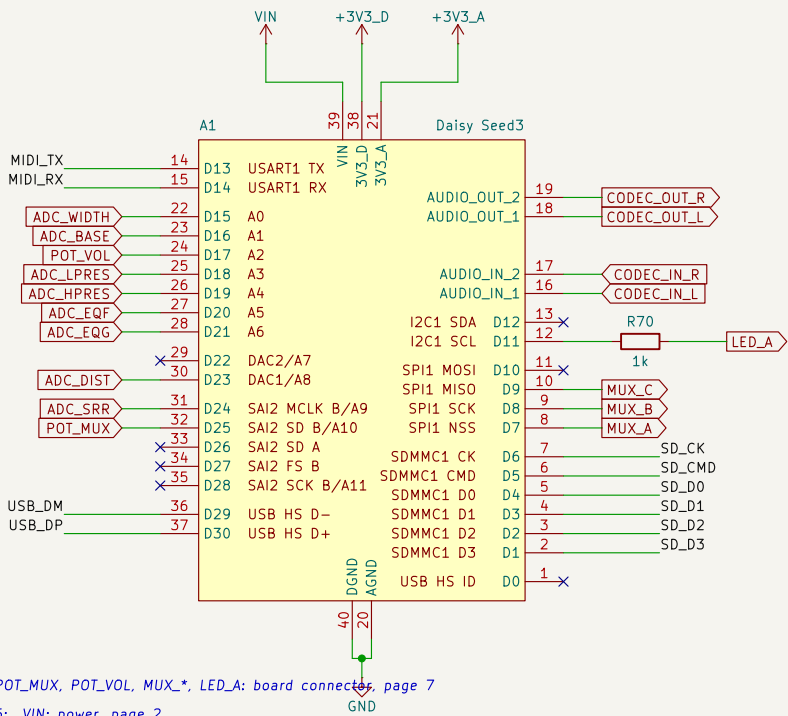
from the Seed3's 3V3_D pin: expansion header J14 (page 3)

−10V_REF

CV input offsets (pages 4–5)

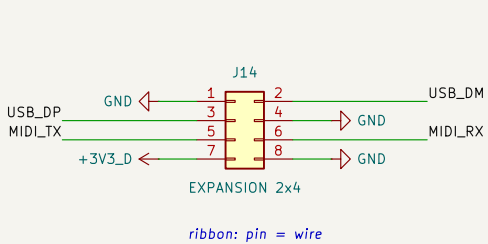
3 Main board: Daisy Seed3, expansion header, clip-LED resistor, optional microSD

Daisy Seed3
on two 1x20 sockets, on the main board's back



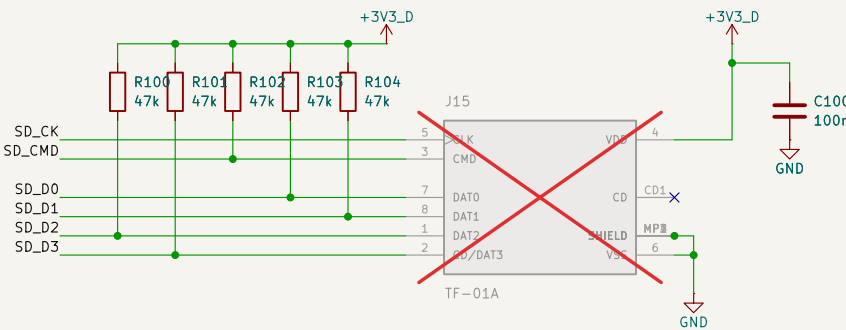
ADC_*: CV inputs, pages 4–5; POT_MUX, POT_VOL, MUX_*, LED_A: board connectors, page 7
CODEC_IN/OUT_*: audio, page 6; VIN: power, page 2
ADC pins, op-amp sections and mux channels follow the floorplan (design/pinmap.py):
the firmware maps them: 22=WIDTH, 23=BASE, 24=POT_VOL, 25=LPRES, 26=HPRES
27=EQF, 28=EQG, 30=DIST, 31=SRR, 32=POT_MUX

Expansion header
to the 2HP USB-C / MIDI expander



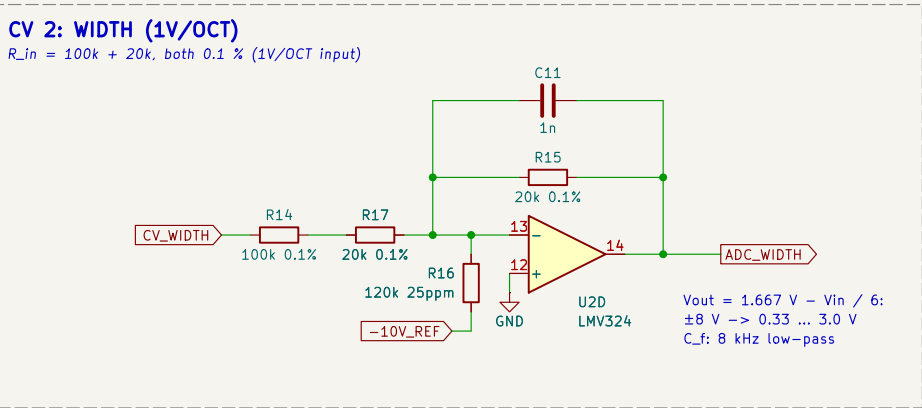
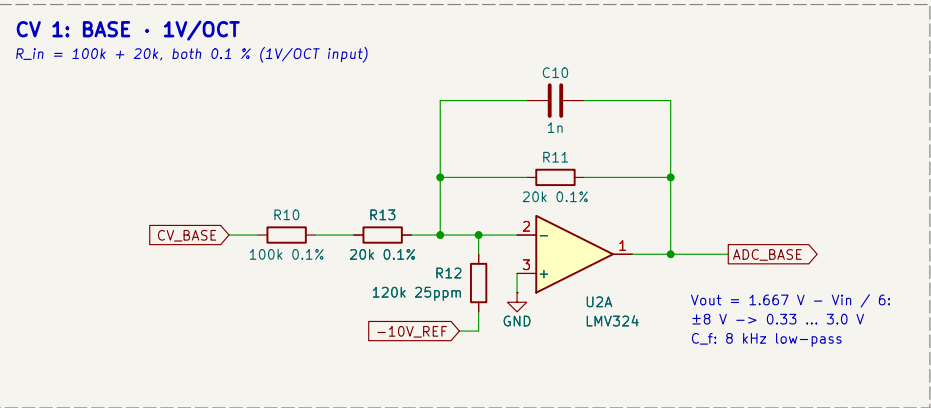
USB_DM/DP, MIDLTX/RX: Seed3 pins 36/37, 14/15 (USART1)

microSD (optional)
J15 DNP: fit per build; the pull-ups are always fitted



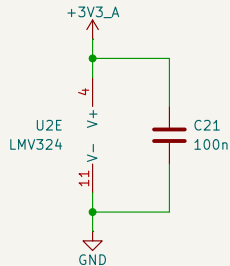
SD_*: Seed3 pins 2–7 (SDMMC1). Firmware enables SD only if the socket is fitted.

4 Main board: CV inputs 1–4, ADC stages

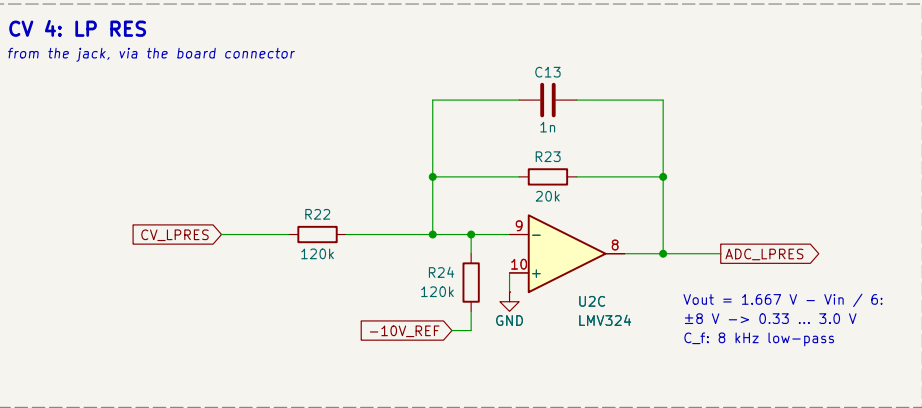
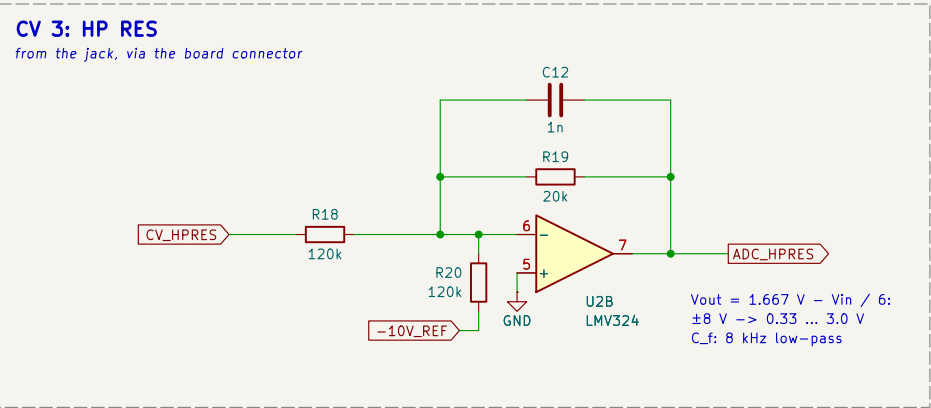


Op-amp supplies

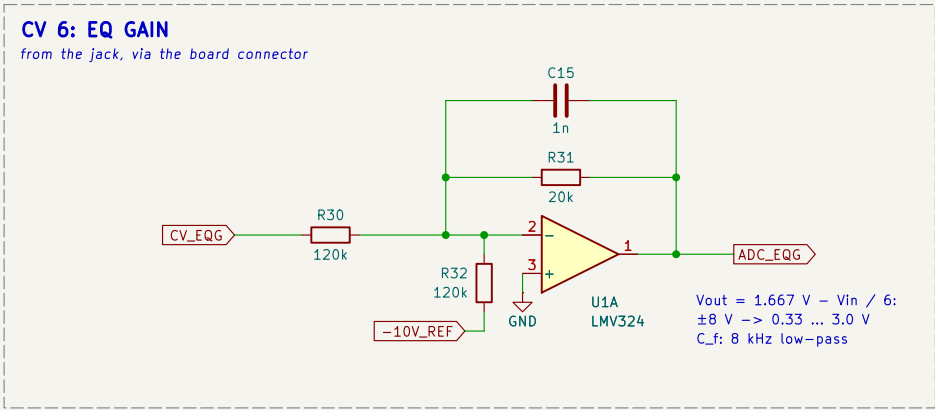
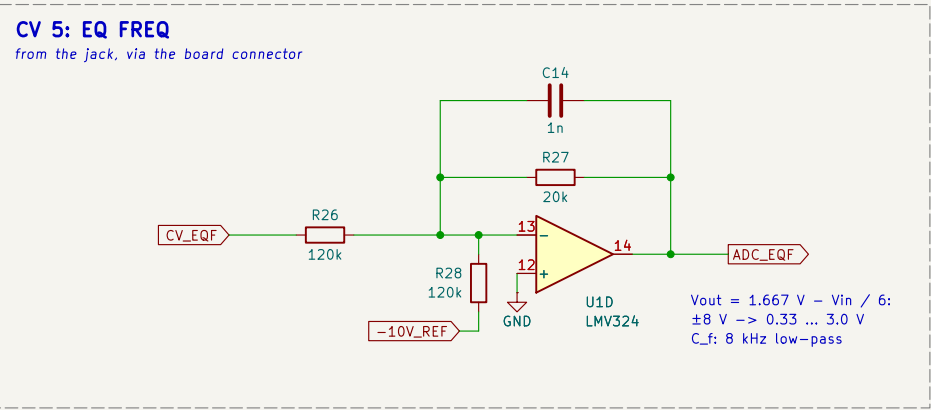
U2: LMV324, ADC stages



Each CV jack (control board) arrives here through the board connector and feeds an inverting stage that scales $\pm 8\text{ V}$ into the Seed's $0 \dots 3.3\text{ V}$ ADC range (ADC_* go to page 3). The same CV drives its LED on the control board. Op-amp sections and ADC pins follow the floorplan (design/pinmap.py).

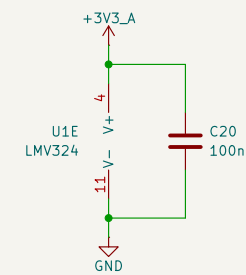


5 Main board: CV inputs 5–8, ADC stages

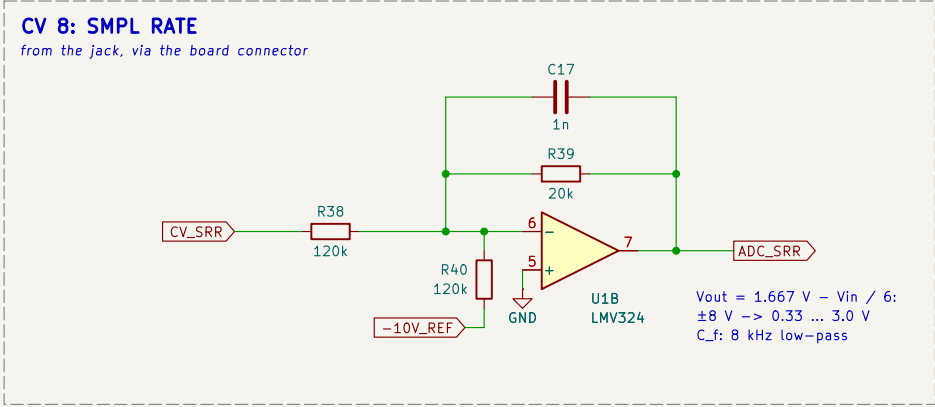
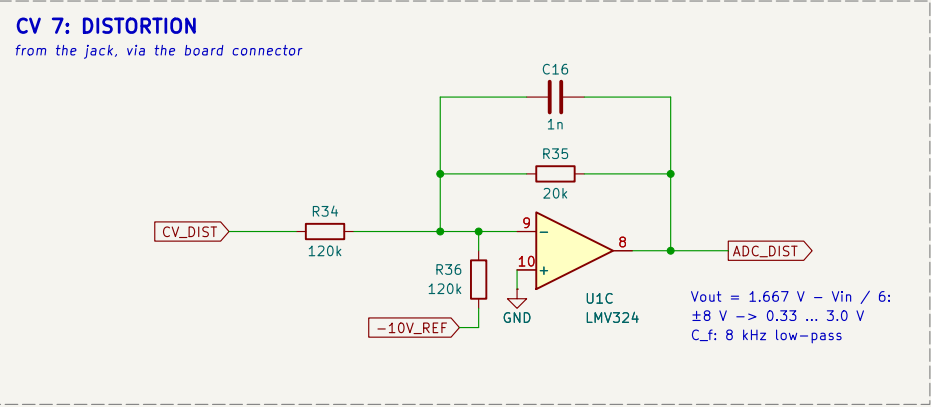


Op-amp supplies

U1: LMV324, ADC stages



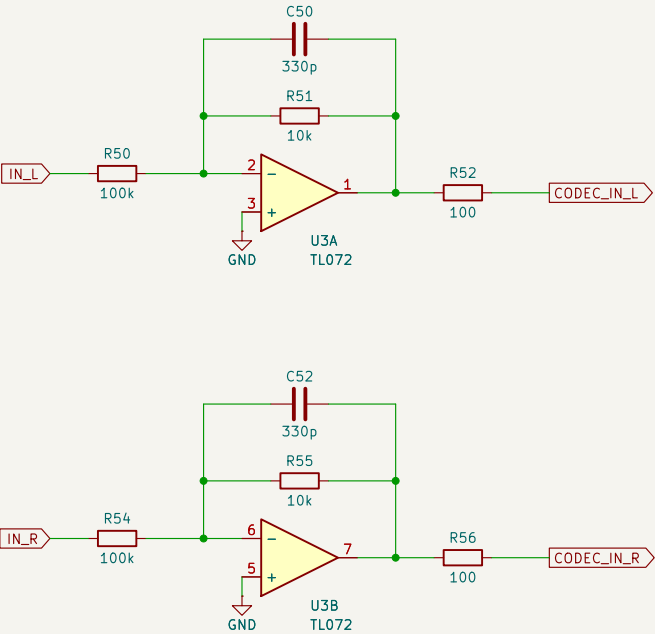
Each CV jack (control board) arrives here through the board connector and feeds an inverting stage that scales $\pm 8\text{ V}$ into the Seed's $0 \dots 3.3\text{ V}$ ADC range (ADC_* go to page 3). The same CV drives its LED on the control board. Op-amp sections and ADC pins follow the floorplan (design/pinmap.py).



6 Main board: audio inputs and outputs

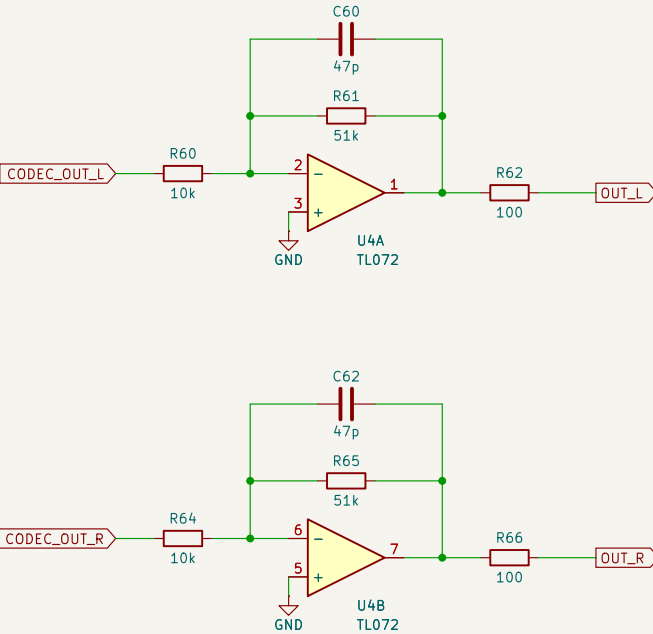
Audio in

gain -0.1 ($100k / 10k$), as the Dev Kit



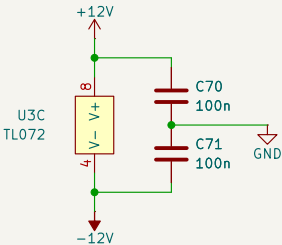
Audio out

gain -5.1 ($10k / 51k$): $0\text{ dBFS} = \pm 7.1\text{ V}$, below the TL072's swing

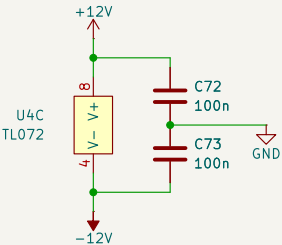


Op-amp supplies

U3: TL072, inputs



U4: TL072, outputs

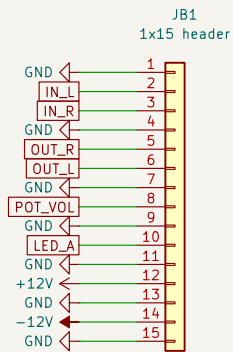


IN_L / IN_R / OUT_L / OUT_R: the audio jacks, on the control board,
through the board connector (page 7). IN R is normal to IN L there.

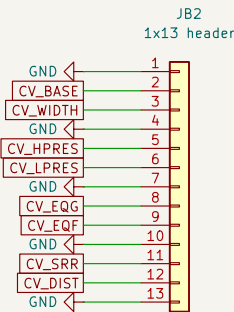
7 Main board: board-to-board connector

Board-to-board connector

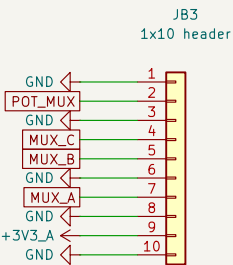
3 single-row 2.54 mm headers: JA* female on the control board's back, JB* male on the main board's front



JB1: pin 1 at (21.95, 69.205) mm,
panel frame; pins run up (main side)



JB2: pin 1 at (41.635, 66.03) mm,
panel frame; pins run down (main side)



JB3: pin 1 at (21.95, 84.445) mm,
panel frame; pins run down (main side)

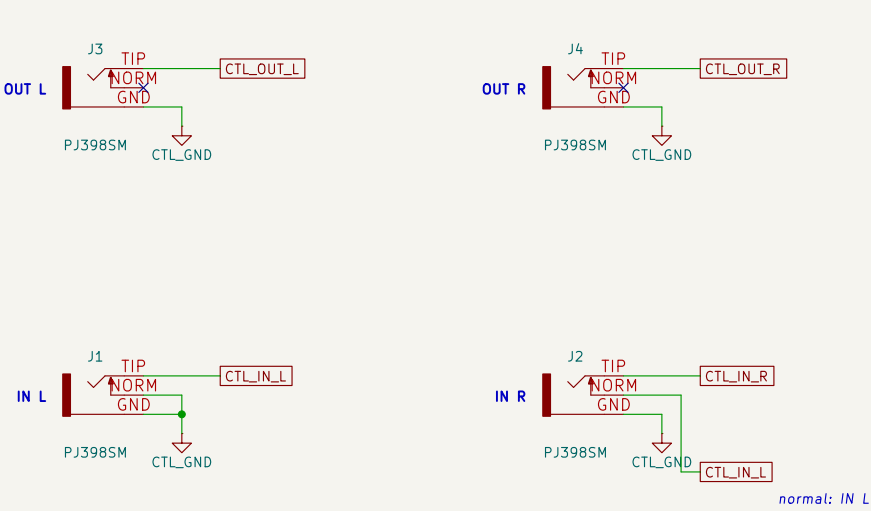
Header groups, positions and pin order come from the floorplan (design/headers.py): analog pairs, then digital pairs, a GND after each pair; supplies between GNDs.

Page 7: Main board: board-to-board connector		
Circuit blocks after Electrosmith's Seed3 Eurorack Dev Kit Rev3 (MIT).		
Drawn by pcb/design/gen_sch.py from boards.py; netlist checked against machine_filter.py (SKiDL).		
MACHINE FILTER (Seed3 build): main board		
Sheet: /Main: connector/		
File: main-connector.kicad_sch		
Title: MACHINE FILTER		
Size: A4	Date: 2026-10-06	Rev: beta
KiCad E.D.A. 10.0.6	Id: 7/12	

8 Control board: jacks and clip LED

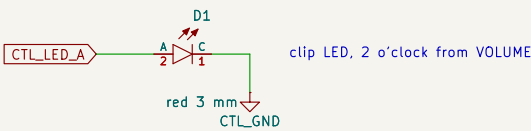
Audio jacks

IN R normalled to IN L; OUT jacks' switch unused



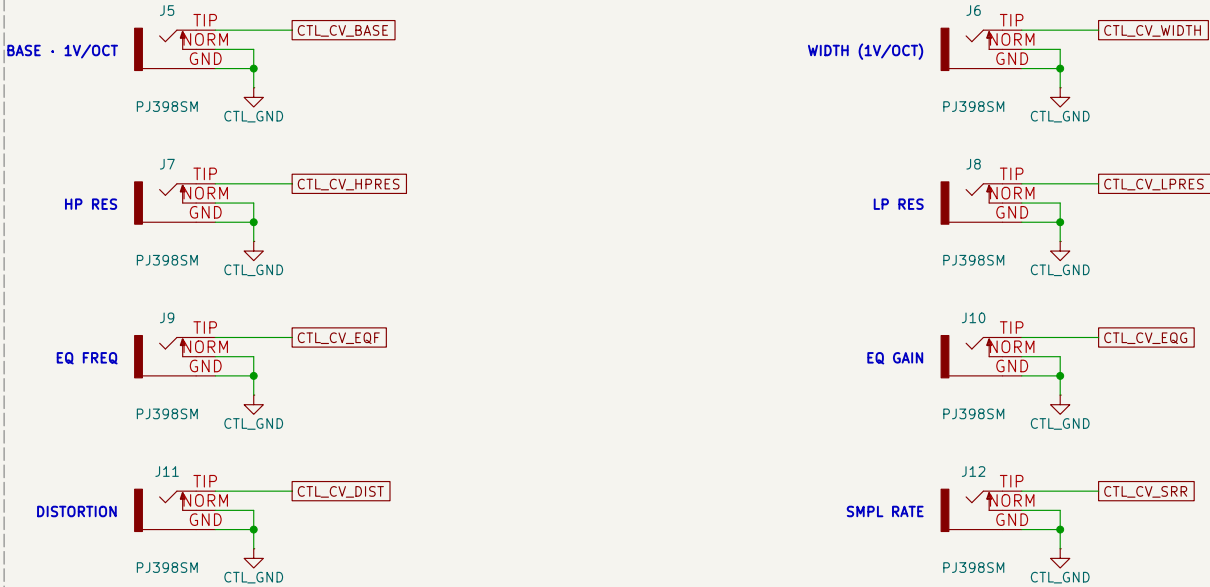
Clip LED

driven from the Seed3 through R70 on the main board



CV jacks

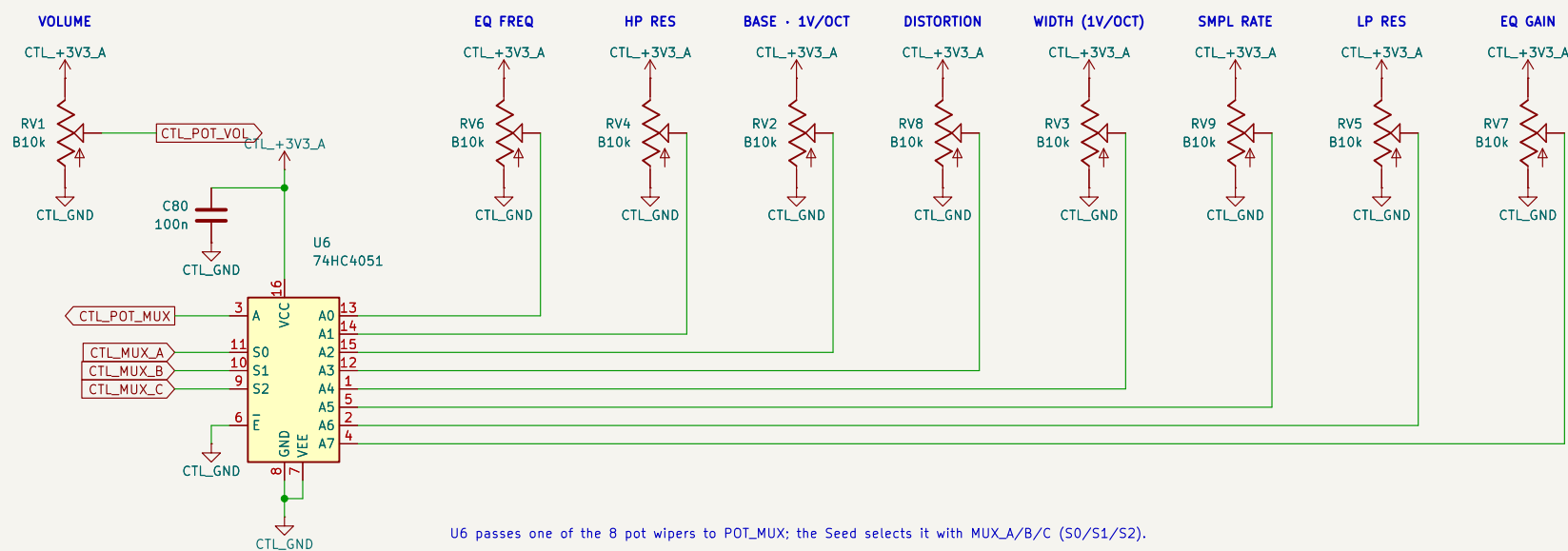
unplugged: the switch grounds the input (0 V, LED dark)



9 Control board: pots and pot multiplexer

Pots and pot multiplexer

pots: CCW end to GND, CW end to +3V3_A (from the main board); VOLUME has its own ADC pin, the other eight share one



U6 passes one of the 8 pot wipers to POT_MUX; the Seed selects it with MUX_A/B/C (S0/S1/S2).

CTL_POT_MUX, CTL_POT_VOL, CTL_MUX_A/B/C go to the Seed3 on the main board, through the board connector (page 12).

Channel order follows the floorplan (design/pinmap.py); the firmware maps it.

Page 9: Control board: pots and pot multiplexer

Circuit blocks after Electrosmith's Seed3 Eurorack Dev Kit Rev3 (MIT).

Drawn by pcb/design/gen_sch.py from boards.py; netlist checked against machine_filter.py (SKiDL).

MACHINE FILTER (Seed3 build): control board

Sheet: /Control: pots/

File: control-pots.kicad_sch

Title: MACHINE FILTER

Size: A4	Date: 2026-10-06
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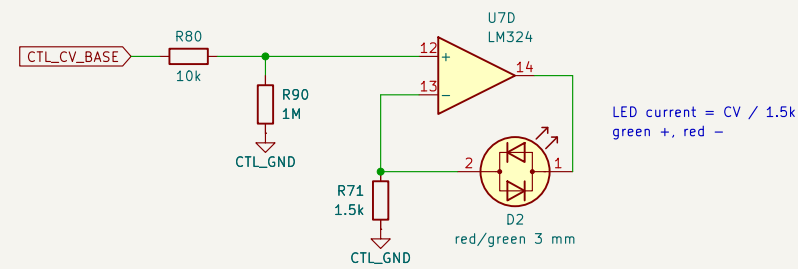
Size: A4	
KiCad E.D.A. 10.0.6	

Rev: beta

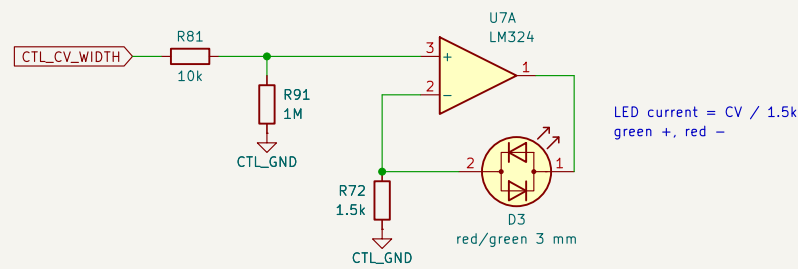
Id: 9/12

10 Control board: CV LEDs 1–4, LED drivers

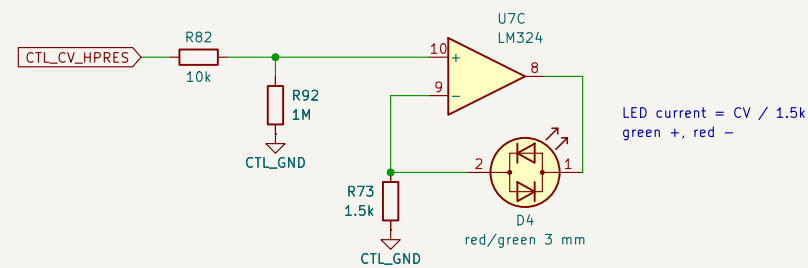
CV 1: BASE · 1V/OCT LED



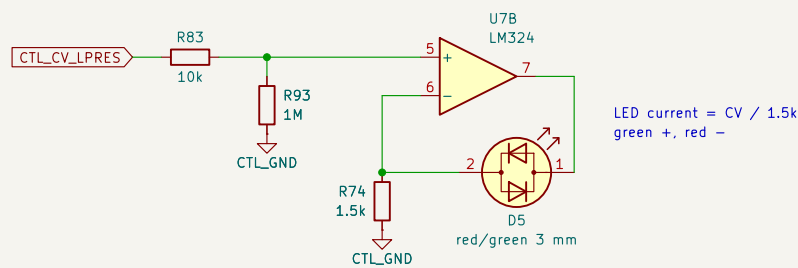
CV 2: WIDTH (1V/OCT) LED



CV 3: HP RES LED

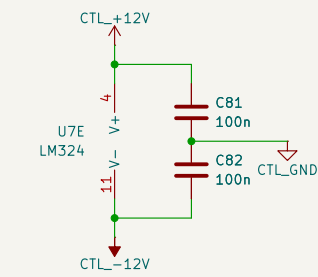


CV 4: LP RES LED



Op-amp supplies

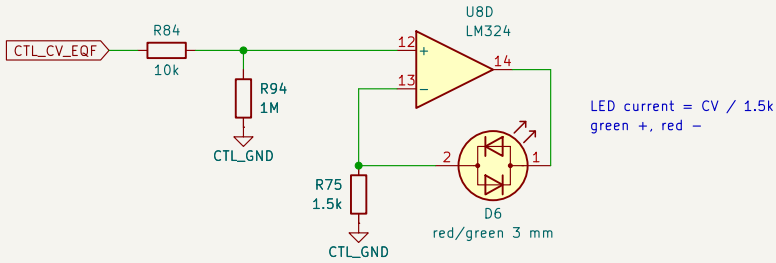
U7: LM324, LED drivers



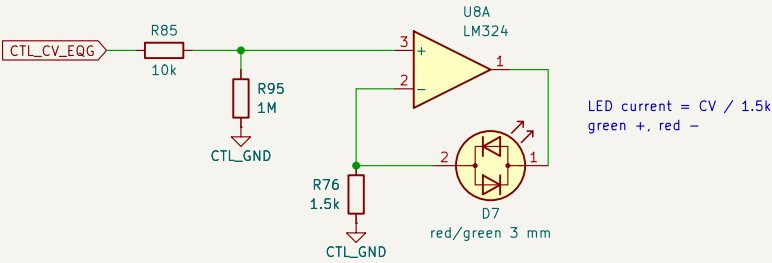
Each CV jack also lights a 2-lead red/green LED by it:
an LM324 with the LED in its feedback loop (after Mutable Instruments' Shades):
LED current = CV / 1.5k, no dead zone, levelling off near 5.5–6 mA from about ±8.5 V.
10k: input protection; 1M: an undriven cable reads 0 V, LED dark.
Through-hole resistors and capacitors (hand-soldered); U7, U8 SOIC (hand-soldered).

11 Control board: CV LEDs 5–8, LED drivers

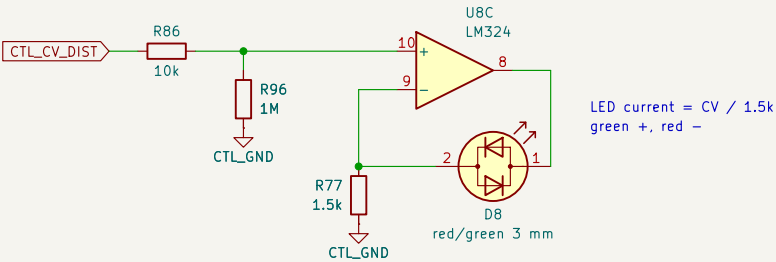
CV 5: EQ FREQ LED



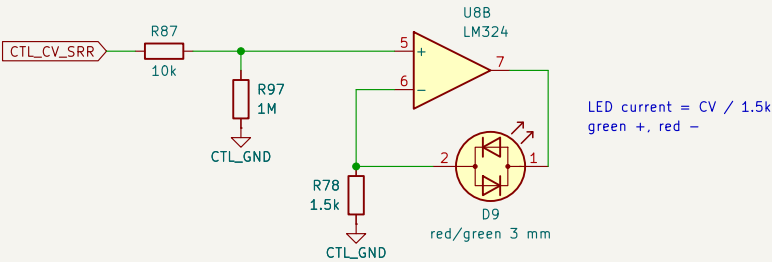
CV 6: EQ GAIN LED



CV 7: DISTORTION LED

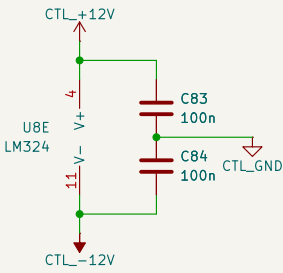


CV 8: SMPL RATE LED



Op-amp supplies

U8: LM324, LED drivers

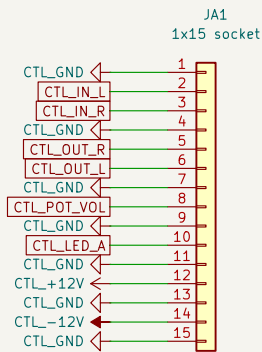


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LED current = CV / 1.5k, no dead zone, levelling off near 5.5–6 mA from about ±8.5 V.
10k: input protection; 1M: an undriven cable reads 0 V, LED dark.
Through-hole resistors and capacitors (hand-soldered); U7, U8 SOIC (hand-soldered).

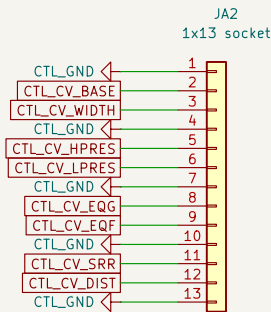
12 Control board: board-to-board connector

Board-to-board connector

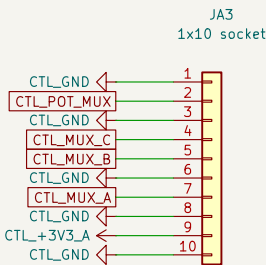
3 single-row 2.54 mm headers: JA* female on the control board's back, JB* male on the main board's front



JA1: pin 1 at (21.95, 69.205) mm,
panel frame; pins run up (main side)



JA2: pin 1 at (41.635, 66.03) mm,
panel frame; pins run down (main side)



JA3: pin 1 at (21.95, 84.445) mm,
panel frame; pins run down (main side)

Supplies from the main board (PWR_FLAG: driven through the connector)



Header groups, positions and pin order come from the floorplan (design/headers.py): analog pairs, then digital pairs, a GND after each pair; supplies between GNDs.

Page 12: Control board: board-to-board connector		
Circuit blocks after Electrosmith's Seed3 Eurorack Dev Kit Rev3 (MIT).		
Drawn by pcb/design/gen_sch.py from boards.py; netlist checked against machine_filter.py (SKIDL).		
MACHINE FILTER (Seed3 build): control board		
Sheet: /Control: connector/ File: control-connector.kicad_sch		
Title: MACHINE FILTER		
Size: A4	Date: 2026-10-06	Rev: beta
KiCad E.D.A. 10.0.6	Id: 12/12	